EXERCISE 8

I Choose the best answer:

1. Which one of the following is not related to random experiment?

   (a) outcomes are predictable in advance

   (b) outcomes is unknown, in advance

   (c) experiment repeatable finite number of times

   (d) experiment is repeatable any number of times.



2. Mathematical probability may also be termed as

   (a) statistical probability (b) classical probability

   (c) Empirical probability (d) None of the above

3. In rolling of a die until 4 appears, the sample space is

   (a) a null set (b) a countable finite set

   (c) a countable infinite set (d) an uncountable set

4. A patient who has undergone a difficult surgery will survive a minimum of 12 years. Then, his survival time, x (in years), can be represented by

   (a) $\{x : 12 \leq x < \infty\}$ (b) $\{x : 12 \leq x < 24\}$

   (c) $\{x : 0 \leq x < 12\}$ (d) any interval along the real line.

5. If A and B are mutually exclusive events them $P(A \cup B)$ is equal to

   (a) $P(A) + P(B)$ (b) $P(A) - P(B)$

   (c) $P(A) + P(B) - P(A \cap B)$ (d) $P(A)P(B)$

6. If $A=\{1, 2\}$; $B=\{3, 4, 5\}$;$C=\{5, 6\}$ are events in S. Then the number sample point in S is

   (a)1 (b)4 (c) 3 (d) 6

7. Probability of not getting 3 when a die is thrown is

   (a) $\frac{1}{3}$ (b) $\frac{5}{6}$ (c) $\frac{1}{6}$ (d) $\frac{1}{4}$

8. If A_1 and A_2 are two events with $P(A_1)=\frac{4}{9}$ and $P(A_2)=\frac{3}{9}$ and $P(A_1 \cap A_2)=\frac{2}{9}$ then the probability that none of these two events occur is

   (a) $\frac{7}{9}$ (b) $\frac{4}{9}$ (c) $\frac{30}{81}$ (d) $\frac{2}{9}$